Numerical Analysis

Chapter 2: Numerical Solutions of Nonlinear Equations of One Variable

Textbook: Classical and Modern Numerical Analysis: Theory, Methods and Practice
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Outline

1. Introduction
2. Bisection method
3. The fixed point method
4. Newton's method
5. Secant method
6. Other topics

Outline

1 Introduction

2 Bisection method

3 The fixed point method

4 Newton's method

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6 Other topics

What's the target problem?

- Solve $f(x) = 0$ for a given function $f(x)$. That is, find the roots of $f(x)$.
- Why do we start from this problem?
- How to guarantee that the given problem has a unique solution on a specified domain?
- How to construct a convergent numerical method?
- How to obtain a faster convergence?
- What if there are more than one roots?
- .....

Reasons to begin with $f(x) = 0$

- This problem occurs frequently.
- The algorithm and the code are easy to develop. The idea and theory are relatively easy to understand and derive. Hence this problem provides us a quick but complete look at different numerical methods for a problem.
- This problem illustrates the iterative method, which is a common numerical technique. It will well prepare us for more complicated iterative methods, not only in our understanding and the theory, but also in coding.
- This problem illustrates that, when many methods are available, selection of the most suitable method depends on the specific problem, the goal, and the computation equipments available.

Existence of the solution

Theorem (I: Intermediate value theorem)

If $f \in C[a,b]$ *and* k *is a number between* $m = \min_{a \le x \le b} f(x)$ *and* $M = \max_{a \le x \le b} f(x)$, *then there exists a number* $c \in [a,b]$ *such that* $f(c) = k$.

Existence of the solution

Corollary (I)

If $f \in C[a,b]$ and $f(a)f(b) < 0$, then there is a $z \in [a,b]$ such that $f(z) = 0$.

Proof.

Consider to apply the intermediate value theorem.

$$f(a)f(b) < 0 \quad \Rightarrow \quad m = \min_{a \le x \le b} f(x) < 0 \text{ and } M = \max_{a \le x \le b} f(x) > 0.$$

Hence $0 \in [m, M]$. Then set the $k = 0$ in the intermediate value theorem, we finish the proof. See Figure 2.1 on page 36 of the textbook for a geometric illustration. □

Uniqueness of the solution

Corollary (II)

If $f \in C[a,b]$, f is monotonic on $[a,b]$ and $f(a)f(b) < 0$, then there is a unique $z \in [a,b]$ such that $f(z) = 0$.

Remark

- *$f'(x)$ can provide the monotonicity of $f(x)$ on certain intervals.*
- *If there are more than one solutions in $[a,b]$, then the numerical methods may or may not converge!*

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Basic idea of the bisection method

- Suppose there is only one solution $z \in [a, b]$.
- Then z must be in either the left half or the right half of the interval $[a, b]$.
- Once we identify the half interval which contains z, we define it to be the new interval and its length is half of that of the original interval $[a, b]$.
- Then we repeat this procedure on the new interval again and again to obtain smaller and smaller intervals which contains the solution z.
- When the interval is small enough, the procedure can be stopped and any point in the interval is a good approximation to the solution z. Usually we pick up the middle point of the last interval to be the final numerical solution.

Algorithm of the bisection method

1. Find a and b such that $f(a)f(b) < 0$ and $f(x)$ is monotonic on $[a, b]$. Otherwise, stop.
2. Initial preparation:
   - (1) Define $a_0 = a$ and $b_0 = b$.
   - (2) Set the iteration counter $k = 0$.
   - (3) Specify a tolerance ε to stop the following loop.
3. While $\frac{b_k - a_k}{2} \geq \varepsilon$, repeat the following procedure in a loop:
   - (1) $x_k = \frac{a_k + b_k}{2}$.
   - (2) If $f(x_k) = 0$, then
     stop the loop;
     elseif $f(a_k)f(x_k) < 0$, then
     $a_{k+1} = a_k$ and $b_{k+1} = x_k$;
     elseif $f(x_k)f(b_k) < 0$, then
     $a_{k+1} = x_k$ and $b_{k+1} = b_k$.
     end
   - (3) $k \leftarrow k + 1$.
4. Output the final x_k from the above loop.

Practice in Matlab for the bisection method

Several implementation issues:

- No need to store all of a_k and b_k. We may just define two variables a and b, then store the current a_k and b_k into them.
- We have two ways to code for a loop: "for" and "while". Example:

for $k = 0 : 10$	*while* $(k < 10)$
.....	.....
end	*end*

- Use "if... elseif...elseif...else...end" structure instead of several "if...end" structures.

Practice in Matlab for the bisection method

- Example 1: $f(x) = e^x + x$, $f(0) = 1$, and $f(-1) = -0.632$. Choose $\varepsilon = 0.01$.
- $f(x)$ is monotonically increasing on any interval since $f'(x) = e^x + 1 > 0$ for all x.
- Let's code for this simple algorithm together!
- Open your Matlab!

Practice in Matlab for the bisection method

Based on the code we developed together just now, we obtain

- Numerical results:

k	a_k	b_k	x_k
0	-1	0	-0.5
1	-1	-0.5	-0.75
2	-0.75	-0.5	-0.625
3	-0.625	-0.5	-0.5625
4	-0.625	-0.5625	-0.59375
5	-0.59375	-0.5625	-0.578125
6	-0.578125	-0.5625	-0.5703125

- Verification: $f(-0.5703125) \approx 4.963760 \times 10^{-3}$. A smaller tolerance ε will give a smaller error.

Practice in Matlab for the bisection method

Several more implementation issues:

- Call a separate subroutine to find a and b such that $f(a)f(b) < 0$. (Independent study problem)
- A tolerance ε_f for $f(x_k)$ can be also defined such that we can stop the loop when $|f(x_k)| < \varepsilon_f$.
- A maximum number M of iteration steps can be also set up to stop the loop in case the iteration does not converge.

Enriched algorithm of the bisection method

1. Call a subroutine to obtain the a and b such that $f(a)f(b) < 0$ and $f(x)$ is monotonic on $[a, b]$. Otherwise, stop.
2. Initial preparation:
   - (1) Define $a_0 = a$ and $b_0 = b$.
   - (2) Set the iteration counter $k = 0$.
   - (3) Specify ε, ε_f and M.
3. While $\frac{b_k - a_k}{2} \geq \varepsilon$ and $k \leq M - 1$, repeat the following procedure:
   - (1) $x_k = \frac{a_k + b_k}{2}$.
   - (2) If $|f(x_k)| < \varepsilon_f$, then
     stop the loop;
     elseif $f(a_k)f(x_k) < 0$, then
     $a_{k+1} = a_k$ and $b_{k+1} = x_k$;
     elseif $f(x_k)f(b_k) < 0$, then
     $a_{k+1} = a_k$ and $b_{k+1} = x_k$.
     end
   - (3) $k \leftarrow k + 1$.
4. Output the final x_k from the above loop.

Convergence theory of the bisection method

Theorem (I)

Assume z is the only solution of $f(x) = 0$ on $[a, b]$. If f is a continuous function on $[a, b]$ and $f(a)f(b) < 0$, then

$$|x_k - z| \leq \frac{b-a}{2^{k+1}}.$$

Proof.

$$z \in [a_k, b_k] \text{ and } \Rightarrow |x_k - z| \leq \frac{b_k - a_k}{2}.$$

Then combine this with $b_k - a_k = \frac{b-a}{2^k}$, we finish the proof. □

Convergence theory of the bisection method

- Example 2: Assume the solution z of $f(x) = 0$ is in $[0, 2]$. How many iteration steps are required to reduce the error to less than 10^{-4}?

- Solution:

$$|x_k - z| \leq \frac{b-a}{2^{k+1}} \leq 10^{-4} \Rightarrow \frac{2-0}{2^{k+1}} \leq 10^{-4}.$$

Then

$$2^k \geq 10^4 \Rightarrow k \geq \frac{ln10^4}{ln2} \approx 13.2877.$$

Hence 14 iteration steps are required.

Convergence theory of the bisection method

Remark

The bisection method is limited since it is not valid for many cases. For example, it is not valid for complex solutions of $f(x) = 0$. *Furthermore, if* $f(x) \geq 0$ *or* $f(x) \leq 0$, *then the bisection method is not valid since* $f(a)f(b) < 0$ *requires either* $f(a) < 0$ *or* $f(b) < 0$.

Remark

Even though the convergence of the bisection method is not fast, the requirements for its convergence is low. Hence the bisection method is a good choice for finding a better real initial point for other advanced iterative numerical methods.

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Basic idea of the fixed point method

- Still consider $f(x) = 0$.
- What if $f(x) = g(x) - x$ or $f(x) = x - g(x)$?
- Then $f(x) = 0 \Rightarrow x = g(x)$.
- Define z to be a fixed point if $z = g(z)$.
- How about this iteration: $x_{k+1} = g(x_k)$?
- A geometric explanation of the fixed point method: see Figure 2.4 on page 46 of the textbook.

Implementation issues

Definition (Cauchy sequence)

A sequence $\{x_k\}_{k=1}^{\infty}$ is called a Cauchy sequence if, given any $\varepsilon > 0$, there exists a N dependent on ε such that $|x_k - x_n| < \varepsilon$ for every k and n greater than N.

Lemma (II)

Cauchy sequences in $\mathbb{R}$ and $\mathbb{C}$ must converge to a point in $\mathbb{R}$ or $\mathbb{C}$.

Implementation issues

- A tolerance ε should be used to stop the loop when $|x_{k+1} - x_k| < \varepsilon$.
- A tolerance ε_f for $g(x_k) - x_k$ can be used to stop the loop when $|g(x_k) - x_k| < \varepsilon_f$.
- A maximum number M of iteration steps can be used to stop the loop in case the iteration does not converge.

Algorithm of the fixed point method

1. Initial preparation:
   - (1) Pick up the initial point x_0.
   - (2) Specify the tolerances ε, ε_f, and the maximum number M of iteration steps.
2. First step: $k = 1$, $x_1 = g(x_0)$.
3. While $|x_k - x_{k-1}| \geq \varepsilon$ and $k \leq M - 1$, repeat the following procedure in a loop:
   - (1) $x_{k+1} = g(x_k)$.
   - (2) If $|g(x_{k+1}) - x_{k+1}| < \varepsilon_f$, then stop the loop.
   - (3) $k \leftarrow k + 1$.
4. Output the final x_{k+1} from the above loop as the the numerical solution.

Practice in Matlab for the fixed point method

- Example 1: $f(x) = \frac{-x+1}{2}$ and $x_0 = 0$. Then $g(x) = \frac{x+1}{2}$. Choose $\varepsilon = 0.01$, $\varepsilon_f = 10^{-6}$, and $M = 100$. It's straightforward to verify that the fixed point is 1.

- Do you feel familiar with the structure of the above algorithm? If so, can you code for it quickly?

- Open your Matlab! Just modify the code of the bisection method a little bit!

Practice in Matlab for the fixed point method

Based on the code we developed together just now, we obtain

- Numerical results:

k	x_k
0	0
1	0.5
2	0.75
3	0.875
4	0.9375
5	0.96875
6	0.984375
7	0.9921875

- Verification:
  $f(0.9921875) = g(0.9921875) - 0.9921875 \approx 0.00390625$. A smaller tolerance ε will give a smaller error.
- Try other initial values? Similar performance.

Practice in Matlab for the fixed point method

- Example 2: $f(x) = 4 + \frac{1}{3}\sin 2x - x$ and $x_0 = 0$. Then $g(x) = 4 + \frac{1}{3}\sin 2x$.
- Numerical results:

k	x_k
0	4
1	4.329786
2	4.230895
3	4.273634
4	4.273634
5	4.256383

- Verification:
  $f(4.256383) = g(4.256383) - 4.256383 \approx 0.00719599.$
- Try other initial values? Similar performance.

Practice in Matlab for the fixed point method

- Example 3: $f(x) = -\frac{x}{2} + \frac{1}{x}$. Then $g(x) = \frac{x}{2} + \frac{1}{x}$. It's straightforward to verify that there are two fixed points: $\sqrt{2}$ and $-\sqrt{2}$. Choose $x_0 = 10$ and $x_0 = -5$ separately.
- Numerical results:

k	x_k from $x_0 = 10$	x_k from $x_0 = -5$
0	10	-5
1	5.1	-2.7
2	2.746078	-1.720370
3	1.737195	-1.441455
4	1.444238	-1.414471
5	1.414526	-1.414214
6	1.414214	

- Verification: $f(1.414214) \approx -4.376268 \times 10^{-7}$ and $f(-1.414214) \approx 4.376268 \times 10^{-7}$.
- Try other initial values? Similar performance.

Practice in Matlab for the fixed point method

- Example 4: $f(x) = x^2 - x$. Then $g(x) = x^2$. Choose $x_0 = 2$ and $x_0 = 0.5$ separately.
- Numerical results:

k	x_k from $x_0 = 2$	x_k from $x_0 = 0.5$
0	2	0.5
1	4	0.25
2	16	0.0625
3	256	3.90625×10^{-3}
4	6.5536×10^4	1.5259×10^{-5}
5	4.2950×10^9	

- Diverge sometimes!

Practice in Matlab for the fixed point method

- Example 5: $f(x) = e^x + x$ and $x_0 = 0$. Then $g(x) = e^x + 2x$ or $g(x) = -e^x$.
- Numerical results:

k	x_k for $g(x) = e^x + 2x$	x_k for $g(x) = -e^x$
0	0	0
1	1	-1
2	4.718282	-0.367879
3	1.214123×10^2	-0.692201
4	5.3539×10^{52}	-0.500474
5		-0.606244
6		-0.545396
7		-0.579612
8		-0.560115
9		-0.571143
10		-0.564879

Practice in Matlab for the fixed point method

- Verification for the case with $g(x) = -e^x$: $f(-0.564879) \approx 3.549922 \times 10^{-3}$.
- Try other initial values for the case with $g(x) = e^x + 2x$?
- Diverge, diverge, diverge for $g(x) = e^x + 2x$! What's wrong?

Practice in Matlab for the fixed point method

- Question 1: When does there exist a solution of the original problem?
- Question 2: When is the solution unique?
- Question 3: When do the numerical solutions $\{x_k\}$ converge to that solution?
- Question 4: How fast do the numerical solutions $\{x_k\}$ converge to that solution?

- Answer: Theory of the conditions for the convergence!

Convergence theory of the fixed point method

Definition (Lipschitz condition)

A function g satisfies a Lipschitz condition on a set G if there exists a Lipschitz constant $L \geq 0$ such that

$$|g(x) - g(y)| \leq L|x - y|, \quad \text{for all } x, y \in G.$$

Furthermore, if g satisfies the Lipschitz condition with $0 \leq L < 1$, then g is said to be a contraction on the set G.

Lemma (I)

If g satisfies a Lipschitz condition on G, then g is continuous on G.

Proof.

Inequality (1) directly gives the continuity based on its definition in the calculus book. □

Convergence theory of the fixed point method

Lemma (II)

Suppose that g is continuously differentiable and that $|g'(x)| \leq L$ for $x \in G$. Then g satisfies a Lipschitz condition with Lipschitz constant L for $x \in G$.

Convergence theory of the fixed point method

Proof.

- Consider $x, y \in G$.
- Without loss of generality, assume that $x < y$. Then

$$\begin{aligned} |g(x) - g(y)| &= \left| \int_x^y g'(s)\, ds \right| \\ &\leq \int_x^y \left| g'(s) \right| \, ds \\ &\leq \int_x^y L \, ds \\ &\leq L \left| x - y \right|. \end{aligned}$$

Convergence theory of the fixed point method

- The above lemma provides a technique to find the Lipschitz constant L.
- Example 6: Suppose $g(x) = -\frac{x^3}{6} + \frac{x^5}{120}$. Find the Lipschitz constant L for g on $G = [-\frac{1}{2}, \frac{1}{2}]$.
- Solution 1: First, $g'(x) = -\frac{x^2}{2} + \frac{x^4}{24}$.

  Since $-\left(\frac{1}{2}\right)^2 \leq -x^2 \leq 0$ and $0 \leq x^4 \leq \left(\frac{1}{2}\right)^4$ on $G = [-\frac{1}{2}, \frac{1}{2}]$, we get $-\frac{1}{8} \leq -\frac{x^2}{2} + \frac{x^4}{24} \leq \frac{1}{384}$.

  Then $|g'(x)| = \left|-\frac{x^2}{2} + \frac{x^4}{24}\right| \leq \frac{1}{8}$. Thus $L = \frac{1}{8}$.

Convergence theory of the fixed point method

- Solution 2: First, $g'(x) = -\frac{x^2}{2} + \frac{x^4}{24}$.

  By completing square, we have $g'(x) = \frac{1}{24}(x^4 - 12x^2) = \frac{1}{24}[(x^2)^2 - 12x^2 + 6^2 - 6^2] = \frac{1}{24}[(x^2 - 6)^2 - 36]$.

  Since $0 \le x^2 \le \frac{1}{4}$ on $G = [-\frac{1}{2}, \frac{1}{2}]$, we get $\frac{1}{24}[(0 - 6)^2 - 36] \le g'(x) \le \frac{1}{24}[(\frac{1}{4} - 6)^2 - 36]$.

  Then $-\frac{47}{384} \le g'(x) \le 0$. Hence $|g'(x)| \le \frac{47}{384}$. Thus $L = \frac{47}{384}$.

Convergence theory of the fixed point method

Theorem (I: Contraction Mapping Theorem in one variable)

Suppose that g satisfies a Lipschitz condition with $0 \leq L < 1$, $g(x) \in G$ for any $x \in G$, and $x_0 \in G$. Then there exists a unique fixed point $z \in G$ such that $z = g(z)$, and the sequence $\{x_k\}$ determined by $x_{k+1} = g(x_k)$ $(k = 0, 1, 2, \cdots)$ converges to z. Moreover, one can obtain the following error estimates:

$$|x_k - z| \leq \frac{L^k}{1-L} |x_1 - x_0|,$$

$$|x_k - z| \leq \frac{L}{1-L} |x_k - x_{k-1}|.$$

Proof.

See pages 40-41 of the textbook or other ways in the literature for the proof. (Independent study problem or take Math 6601 (Numerical Analysis))

Convergence theory of the fixed point method

Remark

This theorem answers Questions 1, 2, and 3.

- *Question 1: When does there exist a solution of the original problem?*
- *Question 2: When is the solution unique?*
- *Question 3: When do the numerical solutions $\{x_k\}$ converge to that solution?*

Convergence theory of the fixed point method

- Recall Question 4: How fast do the numerical solutions $\{x_k\}$ converge to that solution?

Definition (Convergence rate/order)

Let $\{x_k\}$ be a sequence convergent to x^*. If there exist constants C and α and an integer N such that $|x_{k+1} - x^*| \leq C\,|x_k - x^*|^\alpha$ for all $k \geq N$, then we say that the rate of convergence is of order at least α. The rate is said to be linear if $\alpha = 1$ with $C < 1$ and quadratic if $\alpha = 2$.

Convergence theory of the fixed point method

Remark

- *From* $|x_{k+1} = g(x_k)|$, $z = g(z)$ *and the Lipschitz condition, we get*

$$|x_{k+1} - z| = |g(x_k) - g(z)| \leq L\,|x_k - z|.$$

- *Hence the fixed point method has at least linear rate of convergence if the conditions of the Contract Mapping Theory are satisfied.*
- *But is it possible to have a higher rate of convergence?*

Convergence theory of the fixed point method

Theorem (IV: Taylor's Expansion)

Suppose that $f \in C^{n+1}[a,b]$ *and* $x_0 \in [a,b]$. *Then for any* $x \in [a,b]$, *we have the following Taylor's expansion of* $f(x)$ *at* x_0:

$$f(x) = P_n(x) + R_n(x),$$

where

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k \\ &= f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \cdots \\ &\quad + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n, \end{aligned}$$

Convergence theory of the fixed point method

Theorem (IV: Taylor's Expansion(Continued))

$$R_n = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x-x_0)^{n+1}$$

for some $\xi \in [x_0, x]$ *(Lagrange form of the remainder)* ,

or

$$R_n = \frac{1}{n!} \int_{x_0}^{x} f^{(n+1)}(s)(x-s)^n \, ds$$

(Integral form of the remainder) .

Convergence theory of the fixed point method

Theorem (V)

Assume that the iterations $x_{k+1} = g(x_k)$ converge to a fixed point z and $g \in C^q(G)$ where G contains z. Furthermore, assume that q is the first positive integer for which $g^{(q)}(z) \neq 0$ and if $q = 1$ then $g'(z) < 1$. Then the sequence $\{x_k\}$ converges to z with order q.

Remark

- *Usually $g'(z) \neq 0$, hence $q = 1$.*
- *It's not easy to find $g^{(q)}(z)$ $(q = 1, 2, 3, \cdots)$ since the solution z is unknown.*

Convergence theory of the fixed point method

Proof:

- Using $x_{k+1} = g(x_k)$ and the Taylor's expansion of $g(x_k)$ at z, for some ξ_k between x_k and z, we have

$$\begin{aligned} |x_{k+1} - z| &= |g(x_k) - z| \\ &= \Big| g(z) + g'(z)(x_k - z) + \cdots \\ &\quad + \frac{1}{(q-1)!} g^{(q-1)}(z)(x_k - z)^{(q-1)} \\ &\quad + \frac{1}{q!} g^{(q)}(\xi_k)(x_k - z)^q - z \Big| \\ &= \left| \frac{1}{q!} g^{(q)}(\xi_k) \right| |x_k - z|^q . \end{aligned}$$

Convergence theory of the fixed point method

Continued proof:

- Hence

$$\begin{aligned} |x_{k+1} - z| &\leq \left(\sup_{x \in G} \left| \frac{1}{q!} g^{(q)}(x) \right| \right) |x_k - z|^q \\ &= C |x_k - z|^q . \end{aligned}$$

  Thus the sequence $\{x_k\}$ converges to z with order q.

- For linear convergence, we need $C < 1$. And the assumption "if $q = 1$ then $g'(z) < 1$" guarantees $C < 1$ when $q = 1$.

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Basic idea and derivation of the Newton's method

- Still consider $f(x) = 0$.
- $f(x)$ could be complicated and nonlinear.
- Any idea to approximate $f(x)$ by using a simpler function?
- Linearization by using Taylor's expansion!

Basic idea and derivation of the Newton's method

- Taylor's expansion:
  $f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(\xi)(x - x_0)^2$
- Then the linearization is $f(x) \approx p(x) = f(x_0) + f'(x_0)(x - x_0)$.
- Replace $f(x)$ by $p(x) = f(x_0) + f'(x_0)(x - x_0)$ in $f(x) = 0$.
  Then $f(x_0) + f'(x_0)(x - x_0) = 0$.
- Hence $x = x_0 - \frac{f(x_0)}{f'(x_0)}$.
- But the error is big since $\frac{1}{2}f''(\xi)(x - x_0)^2$ could be big.
- Is there anyway to reduce the error?
- Yes, iteration $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$!

Basic idea and derivation of the Newton's method

Another explanation:

- Use the tangent line of $f(x)$ at x_0 to approximate $f(x)$: $y - f(x_0) = f'(x_0)(x - x_0) \Rightarrow y = f(x_0) + f'(x_0)(x - x_0)$.
- Then use the root of the tangent line to approximate the that of $f(x)$. We solve $y = 0$ to obtain: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$.
- Repeat the same procedure for $x_1, x_2, x_3, \cdots$. That is, after we obtain x_k, we use the tangent line of $f(x)$ at x_k to approximate the curve of $f(x)$. Then use the root of the tangent line to approximate the that of $f(x)$.
- Hence we get $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$!

Basic idea and derivation of the Newton's method

Remark

- *A geometric explanation of the Newton's method: see Figure 2.5 on page 49 of the textbook.*
- *This is a special fixed-point method with* $g(x) = x - \frac{f(x)}{f'(x)}$.
- *It's dangerous when* $f'(x_k)$ *is close to* 0.

Basic idea and derivation of the Newton's method

Remark

- *The Newton's method is quadratically convergent: Suppose z is the fixed point, then $g(z) = z$ and $f(z) = 0$. If $f'(z) \neq 0$, then*

$$g'(x) = 1 - \frac{[f'(x)]^2 - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2}$$

$$\Rightarrow \quad g'(z) = \frac{f(z)f''(z)}{[f'(z)]^2} = 0.$$

But $g''(z)$ may not be zero. Thus we can obtain the quadratic convergence rate based on the theorem (V).

- *The quadratic convergence rate also matches the second order remainder $\frac{1}{2}f''(\xi)(x - x_0)^2$ in the above Talyor's expansion.*

Implementation issues

- A tolerance ε should be used to stop the loop when $|x_{k+1} - x_k| < \varepsilon$.
- A tolerance ε_f for $f(x_k)$ can be used to stop the loop when $|f(x_k)| < \varepsilon_f$.
- A maximum number M of iteration steps can be used to stop the loop in case the iteration does not converge.
- A tolerance $\varepsilon_{f'}$ for $f'(x_k)$ can be used to stop the loop when $|f'(x_k)| < \varepsilon_{f'}$.

Algorithm of the Newton's method

1. Initial preparation:
   - (1) Pick up the initial point x_0.
   - (2) Specify the tolerances ε, ε_f, $\varepsilon_{f'}$, and the maximum number M of iteration steps.
2. First step: $k = 0$, $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$.
3. While $|x_{k+1} - x_k| \geq \varepsilon$ and $k \leq M - 1$, repeat the following procedure in a loop:
   - (1) $k \leftarrow k + 1$.
   - (2) $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$.
   - (3) If $|f(x_{k+1})| < \varepsilon_f$ or $|f'(x_{k+1})| < \varepsilon_{f'}$, then stop the loop.
4. Output the final x_{k+1} from the above loop as the the numerical solution.

Practice in Matlab for the Newton's method

- Example 1: $f(x) = e^x + x$. Choose $x_0 = 0$, $\varepsilon = 0.01$, $\varepsilon_f = 10^{-8}$, $\varepsilon_f = 10^{-8}$, and $M = 100$.
- Let's just modify the old code for this new algorithm together!
- Open your Matlab!

Practice in Matlab for the Newton's method

Based on the code we developed together just now, we obtain

- Numerical results:

k	x_k
0	0
1	-0.5
2	-0.566311
3	-0.567143

- Verification: $f(-0.567143) \approx 4.551138 \times 10^{-7}$.

Practice in Matlab for the Newton's method

Comparison among the Newton's method, the bisection method and the fixed point method:

- Verification for the Newton's method after 3 iterations: $f(-0.567143) \approx 4.551138 \times 10^{-7}$.
- Verification for the bisection method after 6 iterations: $f(-0.5703125) \approx 4.963760 \times 10^{-3}$.
- Verification for the fixed point method after 10 iterations: $f(-0.564879) \approx 3.549922 \times 10^{-3}$.

Convergence theory of the Newton's method

Theorem (I)

Let $G \subset \mathbb{R}$. Assume that $f \in C^2(G)$ and f satisfies $|f'(x)| \geq m$ and $|f''(x)| \leq M$ for $x \in G$ where $m, M > 0$. Then for each zero $z \in G$, there is a neighborhood $K_\rho(z) = [z - \rho, z + \rho] \subseteq G$ such that z is the only zero of f in $K_\rho(z)$, and for each $x_0 \in K_\rho(z)$, the approximations x_k $(k = 1, 2 \cdots)$ remain in $K_\rho(z)$ and converge to z with the following error estimates:

$$(a)\ |x_k - z| \leq \frac{2m}{M} q^{2^k}, \quad \text{where } q = \frac{M}{2m} |x_0 - z|,$$

$$(b)\ |x_k - z| \leq \frac{1}{m} f(x_k) \leq \frac{M}{2m} |x_k - x_{k-1}|^2,$$

$$(c)\ |x_{k+1} - z| \leq \frac{M}{2m} |x_k - z|^2.$$

Convergence theory of the Newton's method

Proof.

See pages 50-53 of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □

Convergence theory of the Newton's method

Theorem (II)

Let $G = [a, b]$. Assume that $f \in C^2(G)$, $f(a)f(b) < 0$, and $f'(x)$ is monotonic and never zero on $[a, b]$. Let c denote the value such that $f(c) = \min |f'(a)|, |f'(b)|$. If $\frac{f(c)}{b-a} \leq f'(c)$, then there exists a unique zero z of $f(x)$, and for any $x_0 \in [a, b]$, the approximations x_k $(k = 1, 2 \cdots)$ converge to z.

Theorem (III)

Let $G = [a, b]$. Assume that $f \in C^2(G)$, $f(a)f(b) < 0$, and $f'(x)$ is monotonic and never zero on $[a, b]$. If x_0 can be chosen such that $f(x_0)f''(x_0) < 0$, then there exists a unique zero z of $f(x)$, and for any $x_0 \in [a, b]$, the approximations x_k $(k = 1, 2 \cdots)$ converge to z.

Outline

1 Introduction

2 Bisection method

3 The fixed point method

4 Newton's method

5 Secant method

6 Other topics

Basic idea and derivation of the secant method

- The Newton's method needs to compute $f'(x_k)$. But in realistic circumstances, it may not be possible to compute $f'(x)$ explicitly.
- This is actually the reason why numerical differentiations have been developed.
- One traditional numerical differentiation is $f'(x_k) \approx \frac{f(x_k)-f(x_{k-1})}{x_k-x_{k-1}}$, which is based on the definition of the derivative: $f'(x) = \lim_{y \to x} \frac{f(x)-f(y)}{x-y}$.
- Then we replace the $f'(x_k)$ in the Newton's iteration by the above numerical differentiation to obtain $x_{k+1} = x_k - \frac{f(x_k)(x_k-x_{k-1})}{f(x_k)-f(x_{k-1})}$.

Basic idea and derivation of the secant method

Remark

- *Compared with the Newton's method, the secant method uses the line determined by x_k and x_{k-1}, not the tangent line of $f(x)$ at x_k, to approximate the curve of $f(x)$. Then we use the root of the line determined by x_k and x_{k-1} to approximate the that of $f(x)$.*
- *A geometric explanation of the secant method: see Figure 2.7 on page 59 of the textbook.*

Basic idea and derivation of the secant method

Remark

- *The secant method is not a fixed-point method. It involves not only x_{k+1} and x_k, but also x_{k-1}.*
- *For the secant method, we need two initial values x_0 and x_1.*
- *But we need only one function evaluation $f(x_k)$ at each iteration step since $f(x_{k-1})$ was computed in the previous iteration step.*

Implementation issues

The implementation issues of the secant method is similar to those of the Newton's method, but we don't need the tolerance for the $f'(x_k)$ any more.

- A tolerance ε should be used to stop the loop when $|x_{k+1} - x_k| < \varepsilon$.
- A tolerance ε_f for $f(x_k)$ can be used to stop the loop when $|f(x_k)| < \varepsilon_f$ instead of $f(x_k) = 0$.
- A maximum number M of iteration steps can be used to stop the loop in case the iteration does not converge.

Algorithm of the secant method

1. Initial preparation:
   - (1) Pick up the initial points x_0 and x_1.
   - (2) Set the iteration counter $k = 0$.
   - (3) Specify the tolerances ε, ε_f, and the maximum number M of iteration steps.
2. While $|x_{k+1} - x_k| \geq \varepsilon$ and $k \leq M - 1$, repeat the following procedure in a loop:
   - (1) $k \leftarrow k + 1$.
   - (2) $x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$.
   - (3) If $|f(x_{k+1})| < \varepsilon_f$, then stop the loop.
3. Output the final x_{k+1} from the above loop as the the numerical solution.

Practice in Matlab for the secant method

- Example 1: $f(x) = e^x + x$. Choose $x_0 = 0$, $x_1 = 1$, $\varepsilon = 0.01$, $\varepsilon_f = 10^{-8}$, and $M = 100$.
- Let's slightly modify the code of the Newton's method to obtain the code of the secant method now!
- Open your Matlab!

Practice in Matlab for the secant method

Based on the code we developed together just now, we obtain

- Numerical results:

k	x_k
0	0
1	1
2	-0.367879
3	-0.498592
4	-0.564573
5	-0.567111

- Verification: $f(-0.567111) \approx 5.060399 \times 10^{-5}$.
- Similar performance to the Newton's method.

Convergence theory of the secant method

Theorem (I)

Let G be a subset of $\mathbb{R}$ containing a zero z of $f(x)$. Assume that $f \in C^2(G)$ and there exists an $M \geq 0$ such that

$$M = \frac{\max_{x \in G} |f''(x)|}{2 \max_{x \in G} |f'(x)|}.$$

Let x_0 and x_1 be two initial guesses to z and let $K_\varepsilon(z) = (z - \varepsilon, z + \varepsilon) \in G$ where $\varepsilon = \frac{\delta}{M}$ and $\delta < 1$. Let $x_0, x_1 \in K_\varepsilon(z)$. Then the approximations x_k $(k = 2, 3 \cdots)$ remain in $K_\varepsilon(z)$ and converge to z with error

$$|x_k - z| \leq \frac{1}{M} \delta^{\left(\frac{1+\sqrt{5}}{2}\right)^k}$$

Convergence theory of the secant method

Proof.

See pages 60-62 of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □

Convergence theory of the secant method

Theorem (II)

Let $G = [a, b]$. Assume that $f \in C^2(G)$, $f(a)f(b) < 0$, and $f'(x)$ is monotonic and never zero on $[a, b]$. Assume x_0 is chosen to be one of a and b and x_1 is chosen to be the other one of a and b. If $f(x_0)f''(x_0) < 0$, then there exists a unique zero z of $f(x)$, and for any $x_0 \in [a, b]$, the approximations x_k $(k = 1, 2 \cdots)$ converge to z with a linear rate.

Outline

1 Introduction

2 Bisection method

3 The fixed point method

4 Newton's method

5 Secant method

6 Other topics

Topics: Independent study problems

- Different convergence theorems?
- How to deal with multiple zeros?
- Different modified Newton's methods: the simplified Newton's method, the Newton's method with a downhill parameter, the interval Newton's method (section 2.5), and so on.

  Solving nonlinear equations with Newton's by C. T. Kelly, Fundamentals of Algorithms, 1. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2003.

Topics: Independent study problems

- The Müller's method(section 2.6.2): Compared with the secant method , the Müller's method uses a parabola, which is determined by x_k, x_{k-1} and , x_{k-2}, to approximate the curve of $f(x)$. Then we use the root of the parabola to approximate the that of $f(x)$.

- Aitken acceleration (section 2.7.1): Combine the formulas of two or more iteration steps to obtain an approximation with higher rate of convergence.

- The Steffensen's method (section 2.7.2).

- Roots of Polynomial (section 2.8).

- Many other methods.....